

Modularidade

‘Evolução de Fenótipos Complexos’

Monique N. Simon

IB – USP

monique.simon@usp.br

Por que estudar variação em biologia?

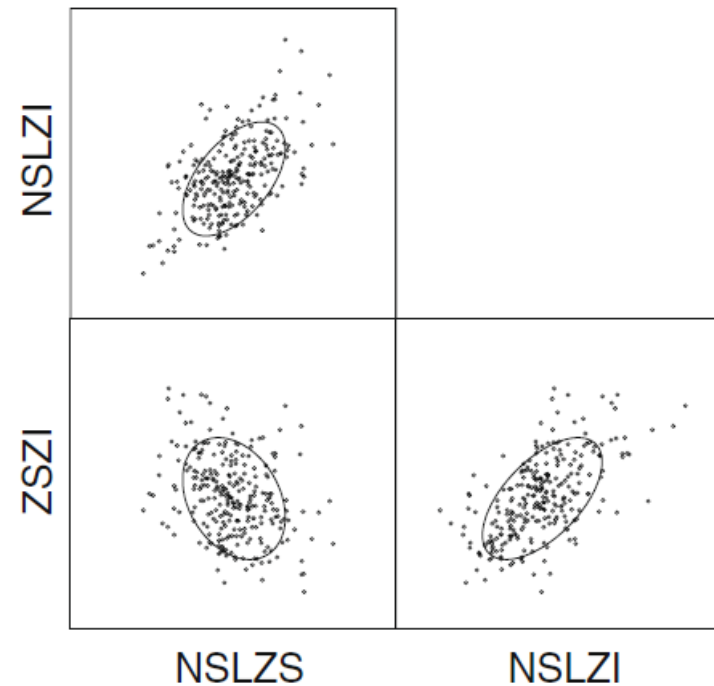
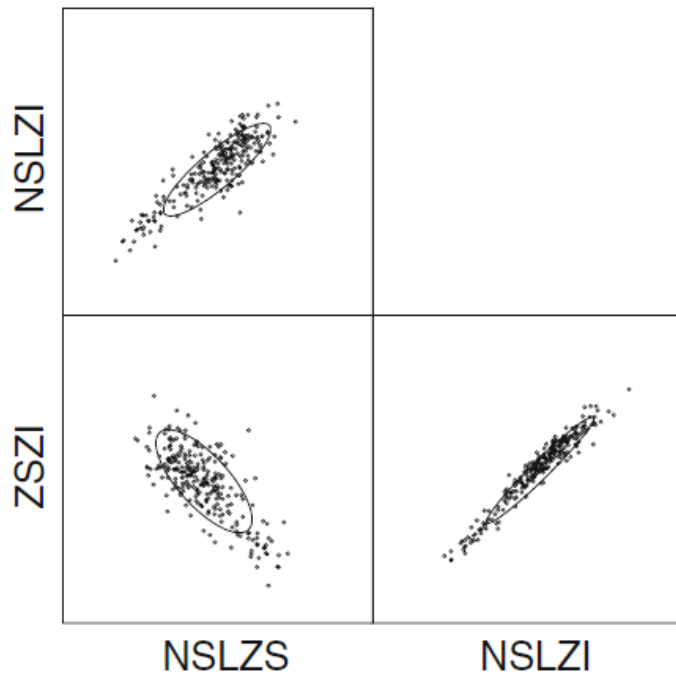
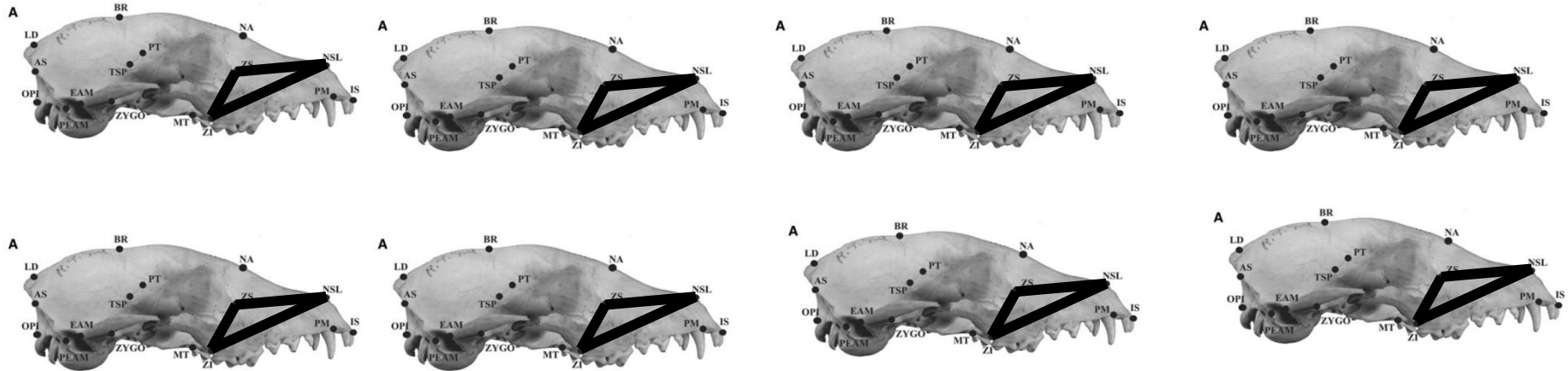
Quais as origens de variação fenotípica e genética?

Como a variação é organizada e representada em populações?

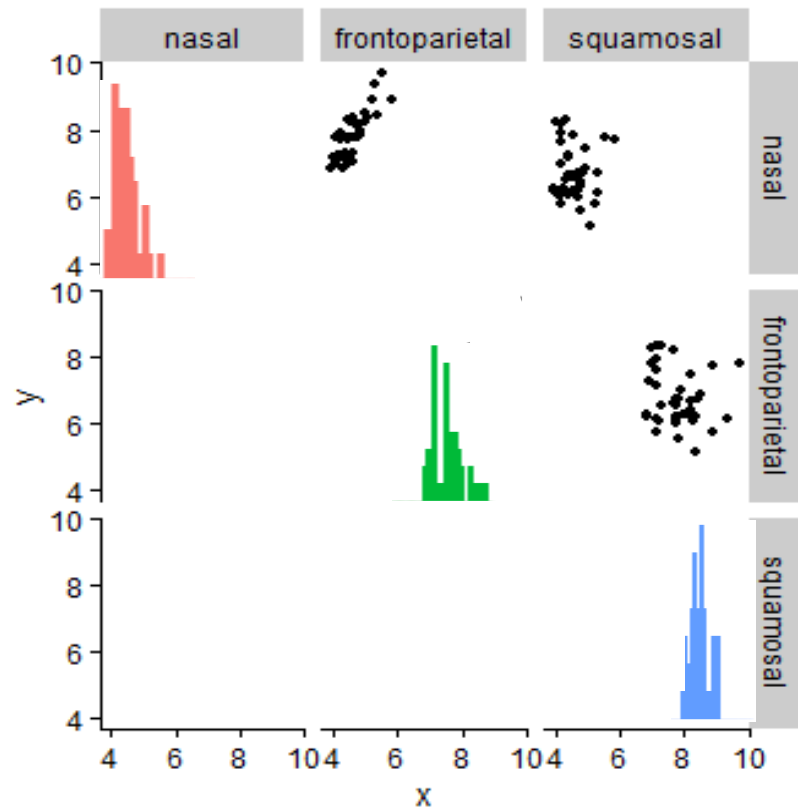
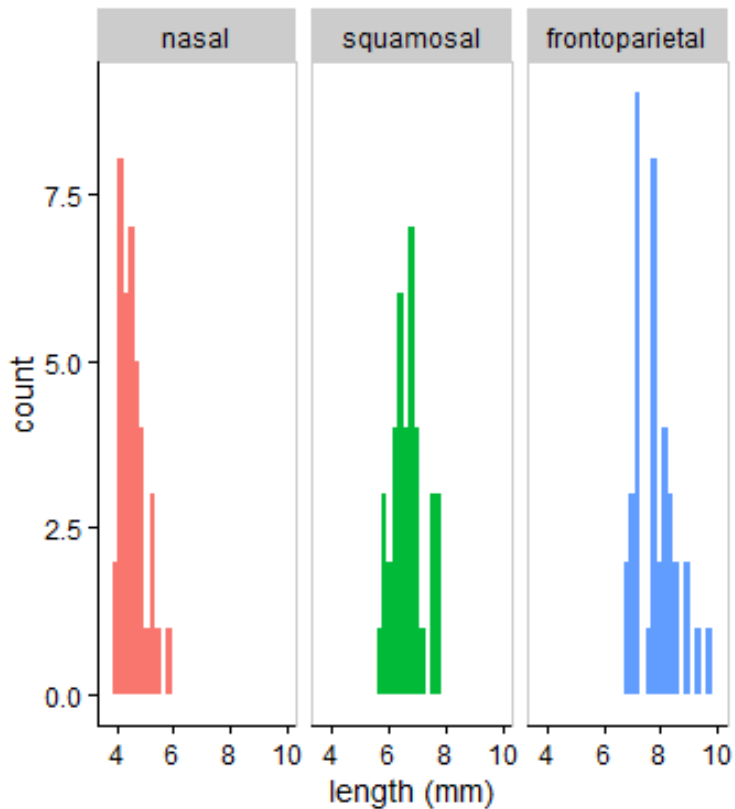
Quais são as consequências evolutivas de como a variação é organizada nas populações?

Múltiplas dimensões: como representar?

Porto et al. 2009



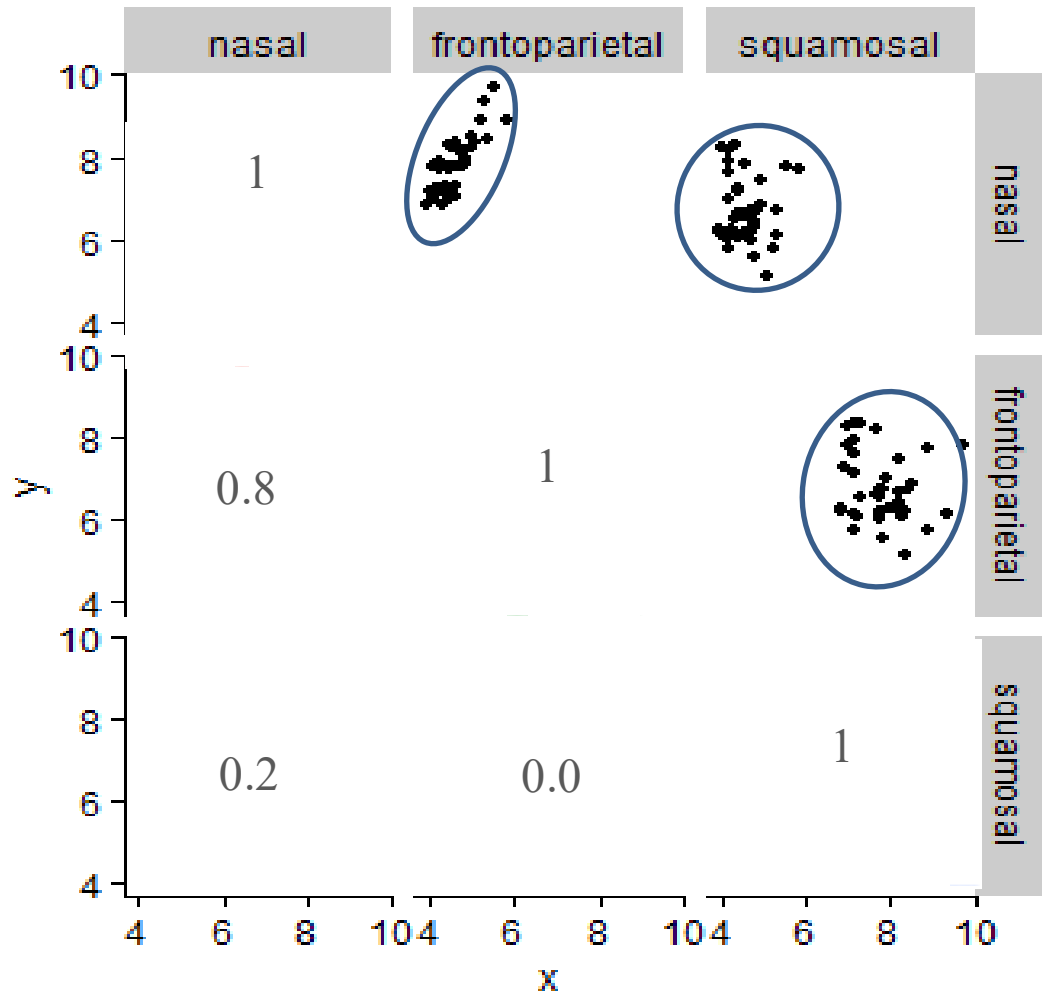
Matriz Fenotípica = matriz de variância e covariância



Tem que haver variação nos valores dos caracteres na população!

$$\text{Cov}(X, Y) = \frac{\sum (X_i - \bar{X})(Y_j - \bar{Y})}{n}$$

Matriz Fenotípica = matriz de correlação



$$r = r_{xy} = \frac{\text{Cov}(x, y)}{S_x \times S_y}$$

Teoria de Integração Morfológica/Modularidade (1958, 1960)



Kenneth P. O'Leary

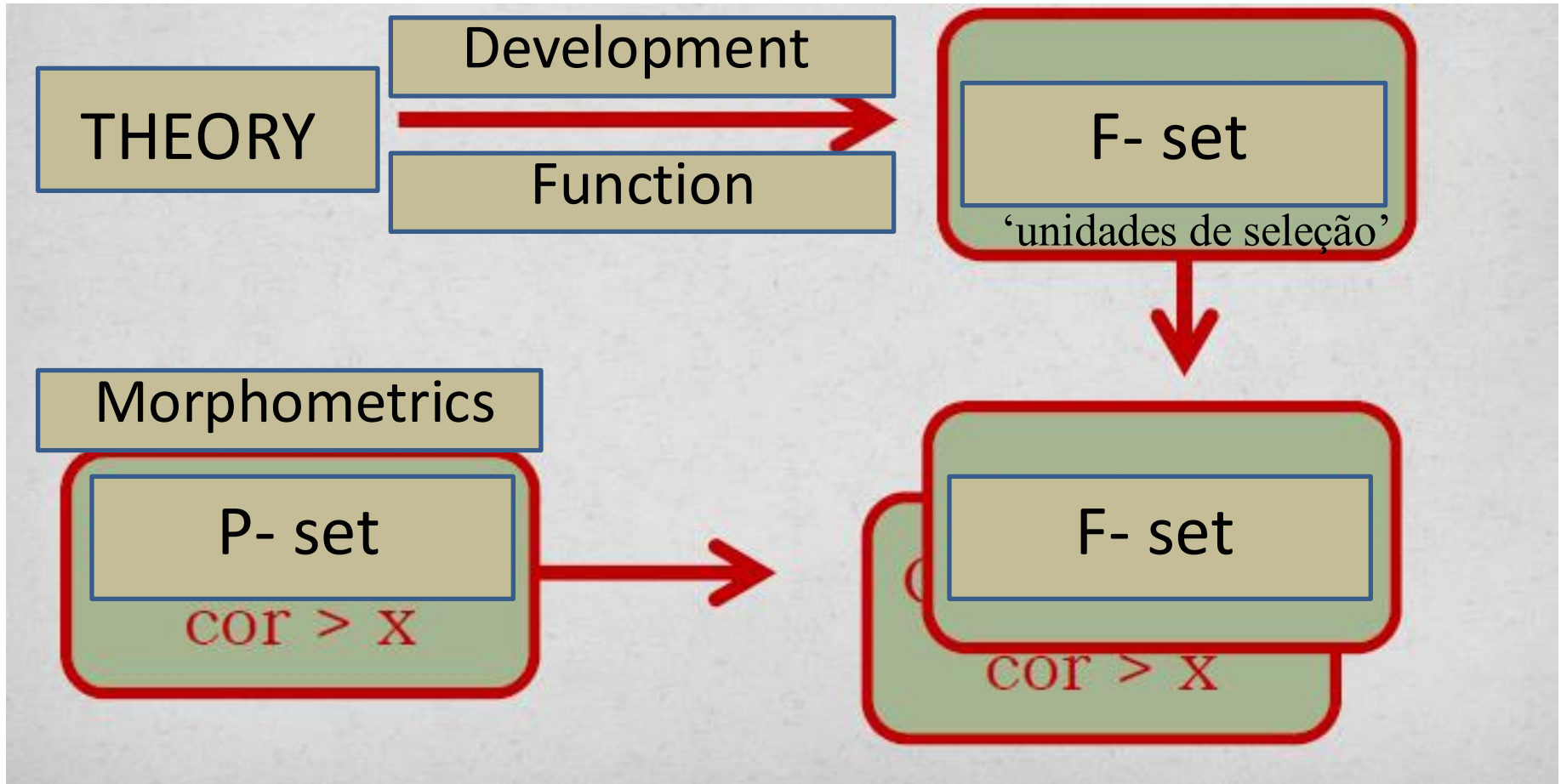


Photo by Ruth O. Hotton

Raissa Berg

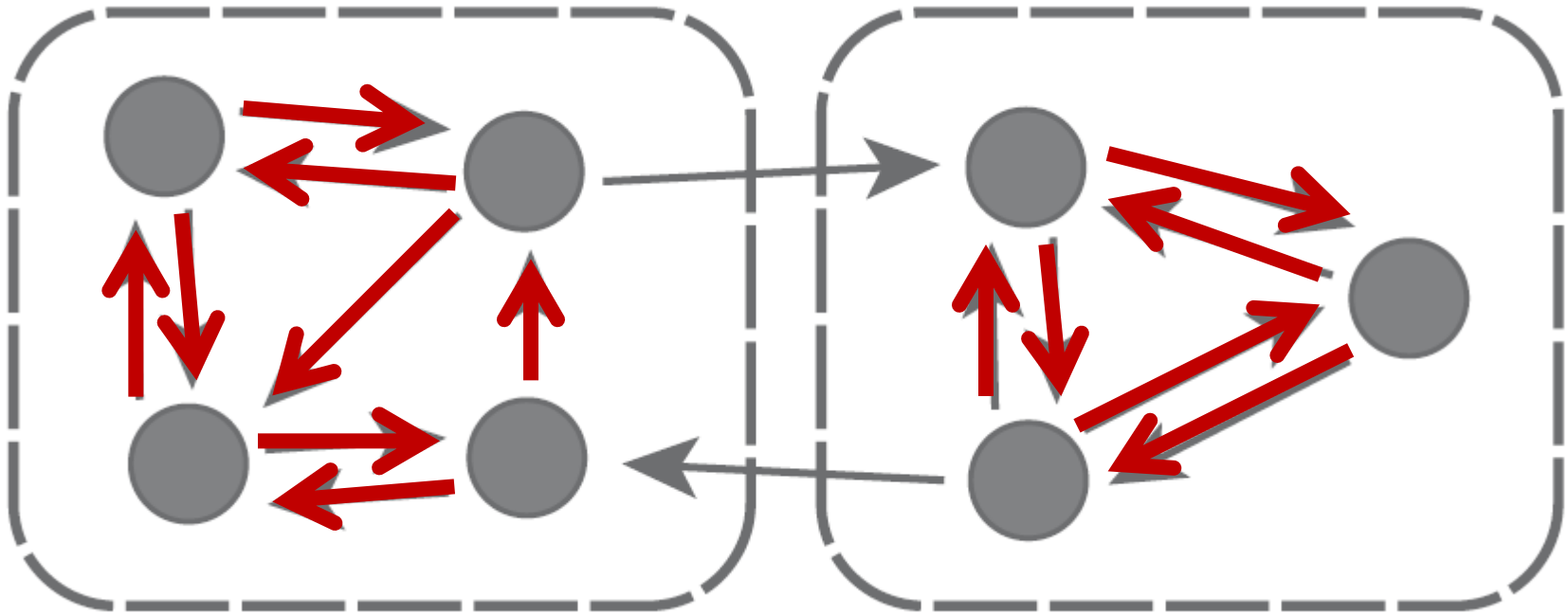
Correlações Fenotípicas = Relações entre caracteres

Olson and Miller (1958)



P-set são medidas estatísticas de covariação!

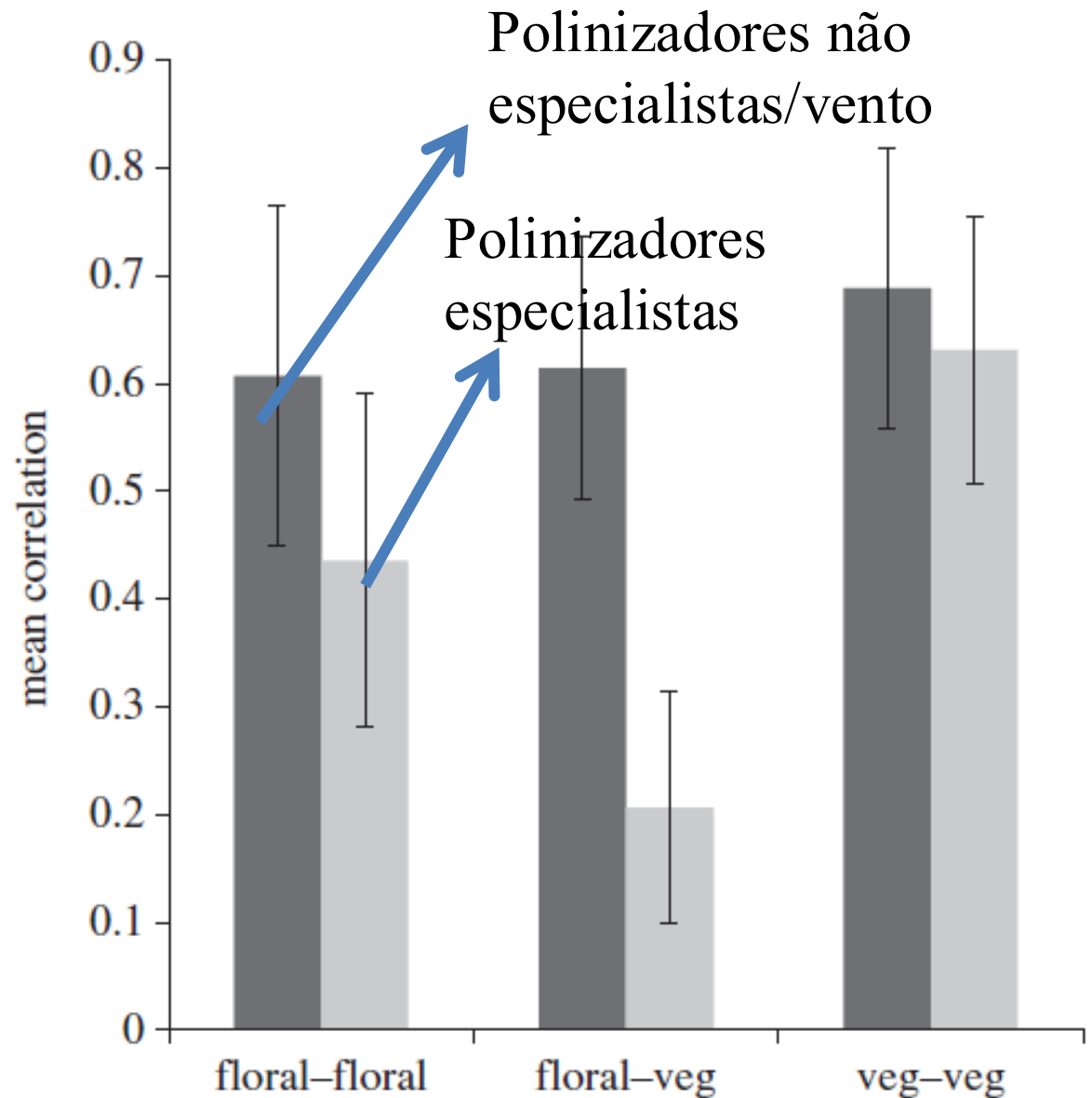
Integração dentro de unidades



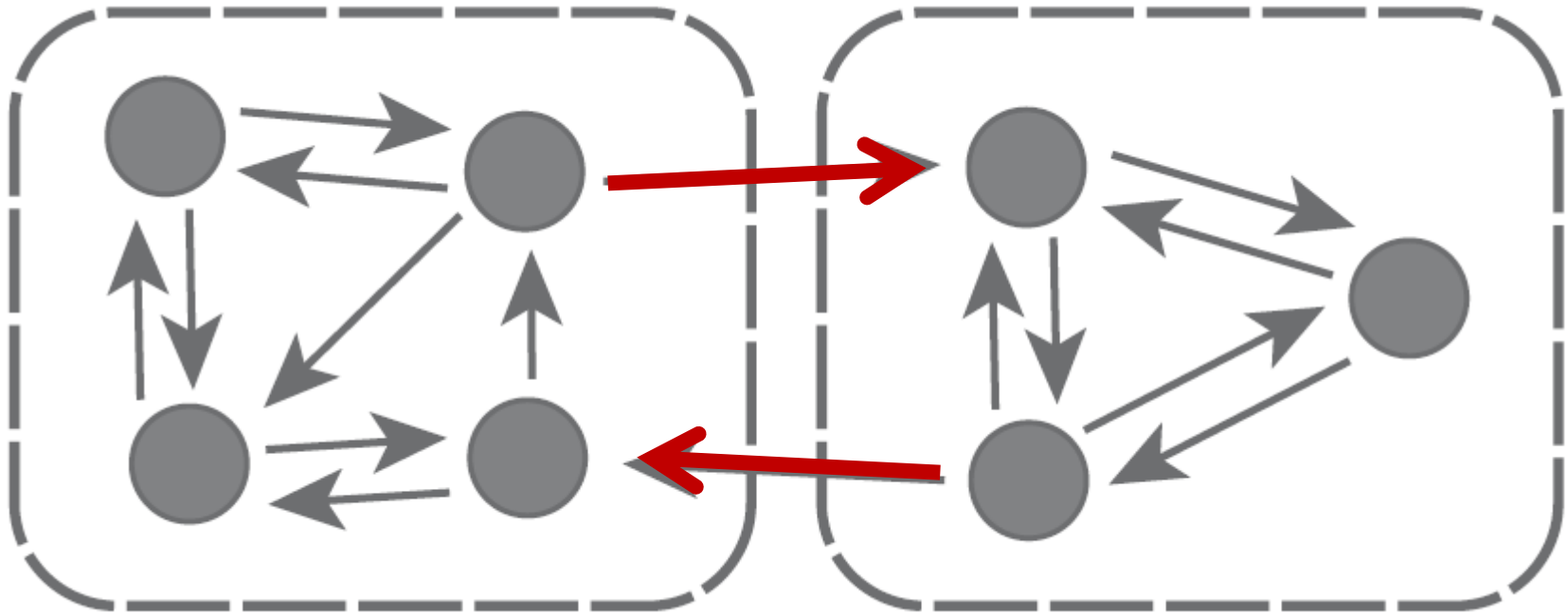
"It seemed evident . . . that character changes occurring in evolution of species **could not be considered to be independent of each other** and that studies which did not consider this dependency ignored a significant aspect of change . . ."

(Olson and Miller, 1958).

Berg e Independência entre plêiades de correlação



Modularidade: Semi-independência das unidades



‘In all the instances of existence of pleiades it was the flower that had become free of the correlation interdependences within the plant organism’.

(Berg 1960)

Cheverud 1982: Cayo rhesus macaques

$N = 51$ mother/offspring pairs



Unidades funcionais

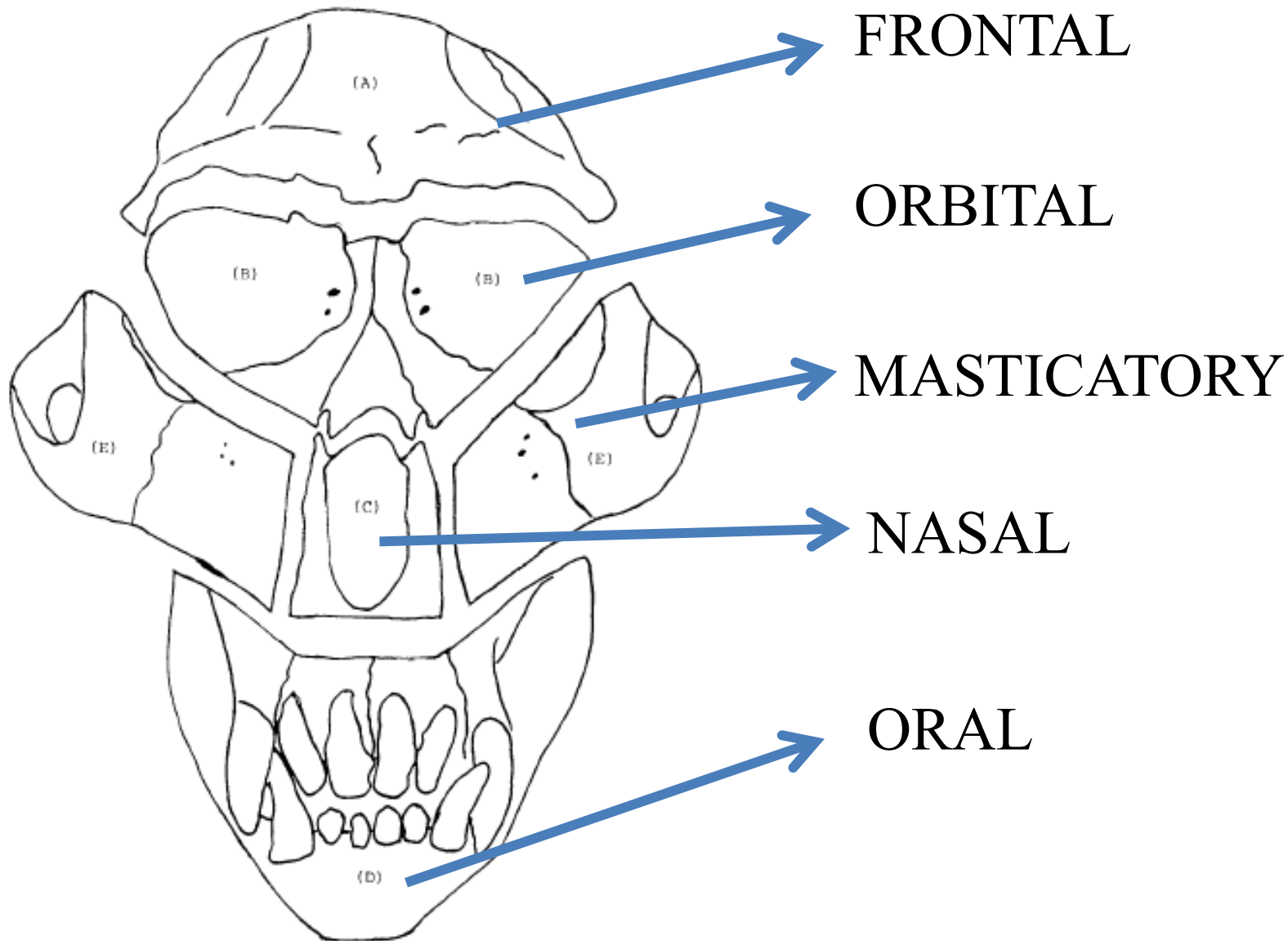
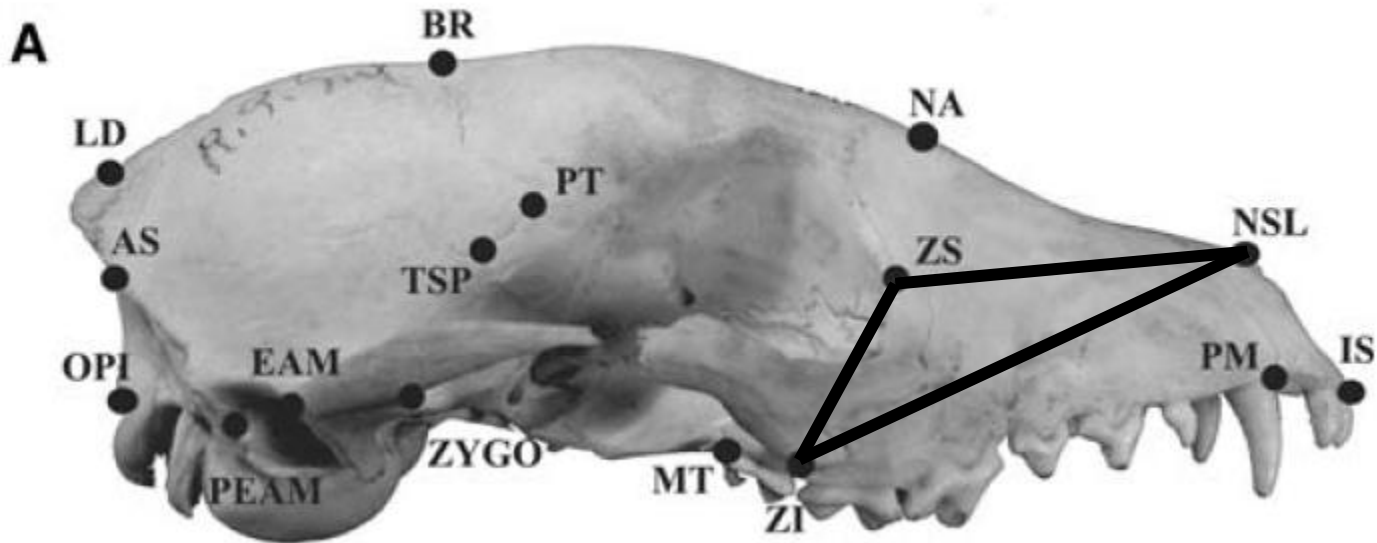
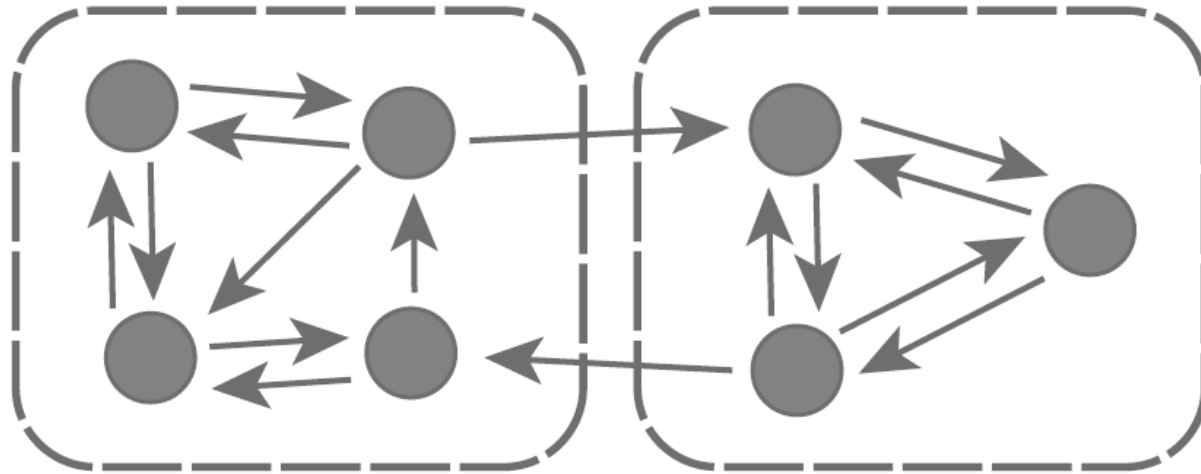


TABLE 5. *Average phenotypic, genetic, and environmental correlations and squared correlations within and between F-sets.*

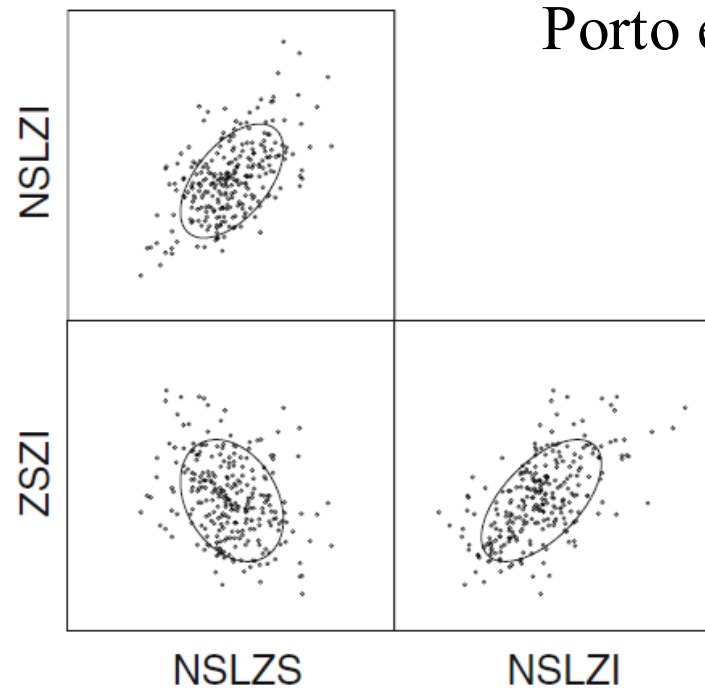
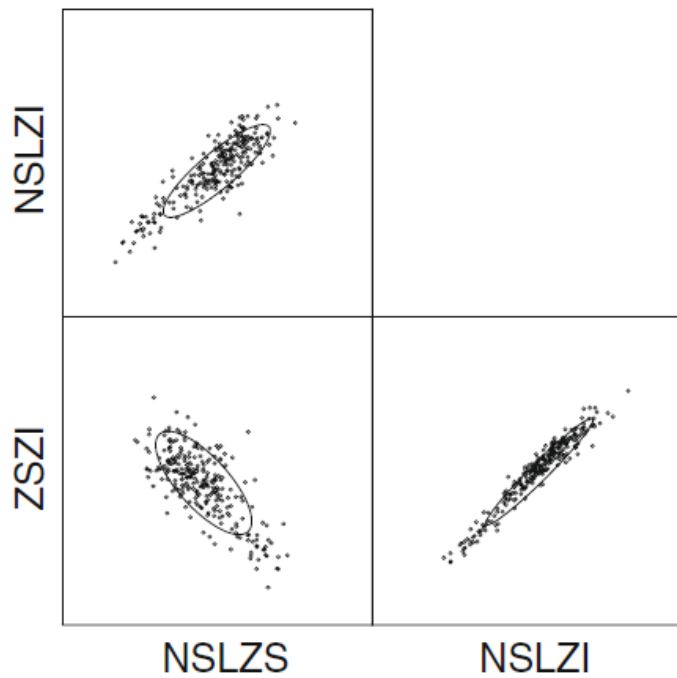
Correlation	Within <i>F</i> -set r	Between <i>F</i> -set r	Within <i>F</i> -set r^2	Between <i>F</i> -set r^2
Phenotypic	0.269	0.105	0.127	0.023
Environmental	0.281	0.093	0.228	0.092
Genetic	0.270	0.138	0.385	0.248
Genetic (neuro.)	0.370	0.110	0.517	0.159
Genetic (facial)	0.123	-0.042	0.190	0.264

‘Individual phenotypic characters and genes only evolve within the larger context of the organism in which they occur. Therefore, **the organisms are the integrated functional units which evolve.**’

Padrão e Magnitude de Integração



Mesmo padrão, diferentes magnitudes de integração



Porto et al. 2009

A	NSLZS	NSLZI	ZSZI
NSLZS	1		
NSLZI	0.85	1	
ZSZI	-0.70	0.90	1

B	NSLZS	NSLZI	ZSZI
NSLZS	1		
NSLZI	0.45	1	
ZSZI	-0.30	0.50	1

Integração e Modularidade

Como se conectam os padrões medidos com os processos subjacentes?

PROCESSOS
INDIVIDUAIS

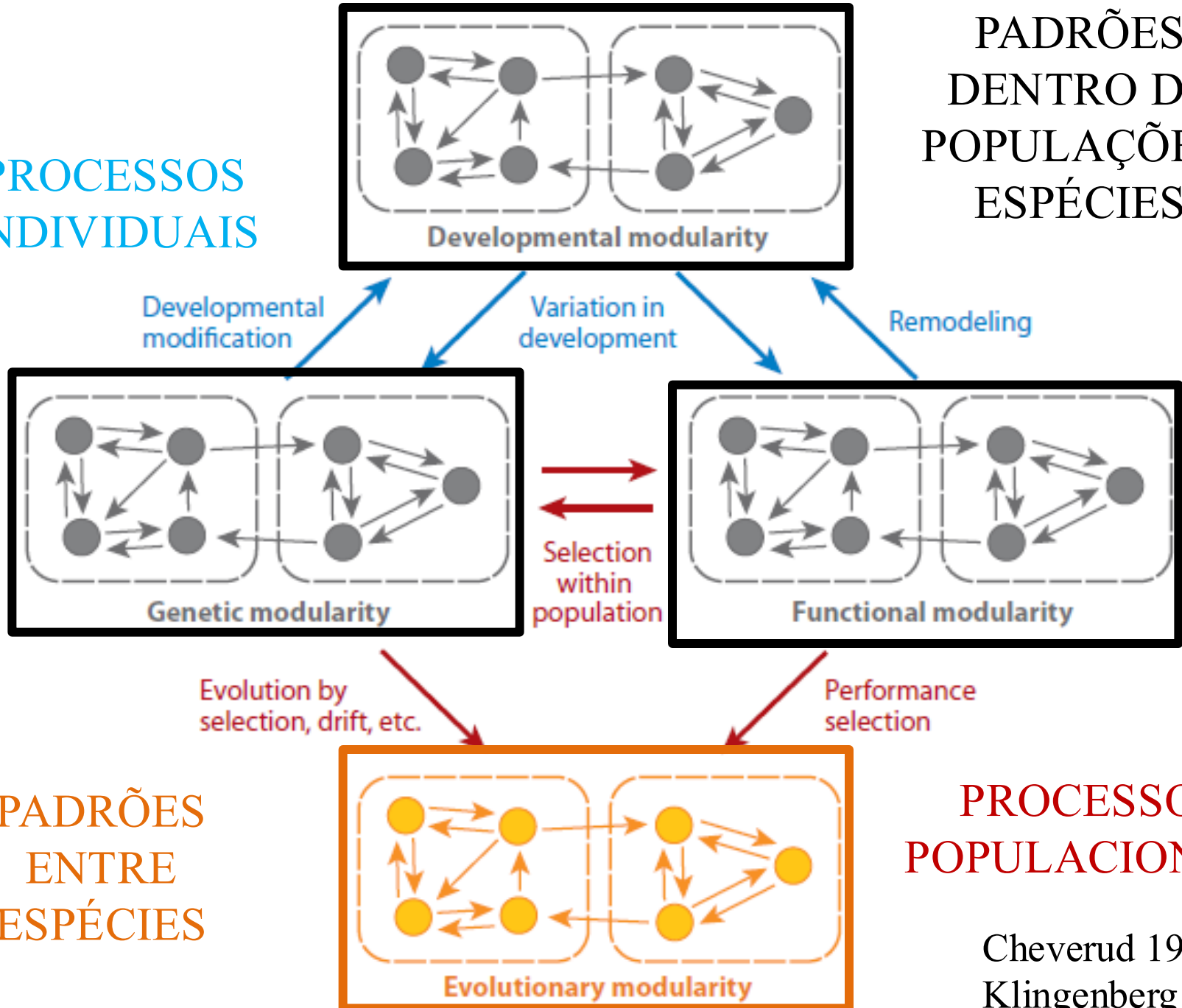
PADRÕES
DENTRO DE
POPULAÇÕES/
ESPÉCIES

PROCESSOS
POPULACIONAIS

PADRÕES
ENTRE
ESPÉCIES

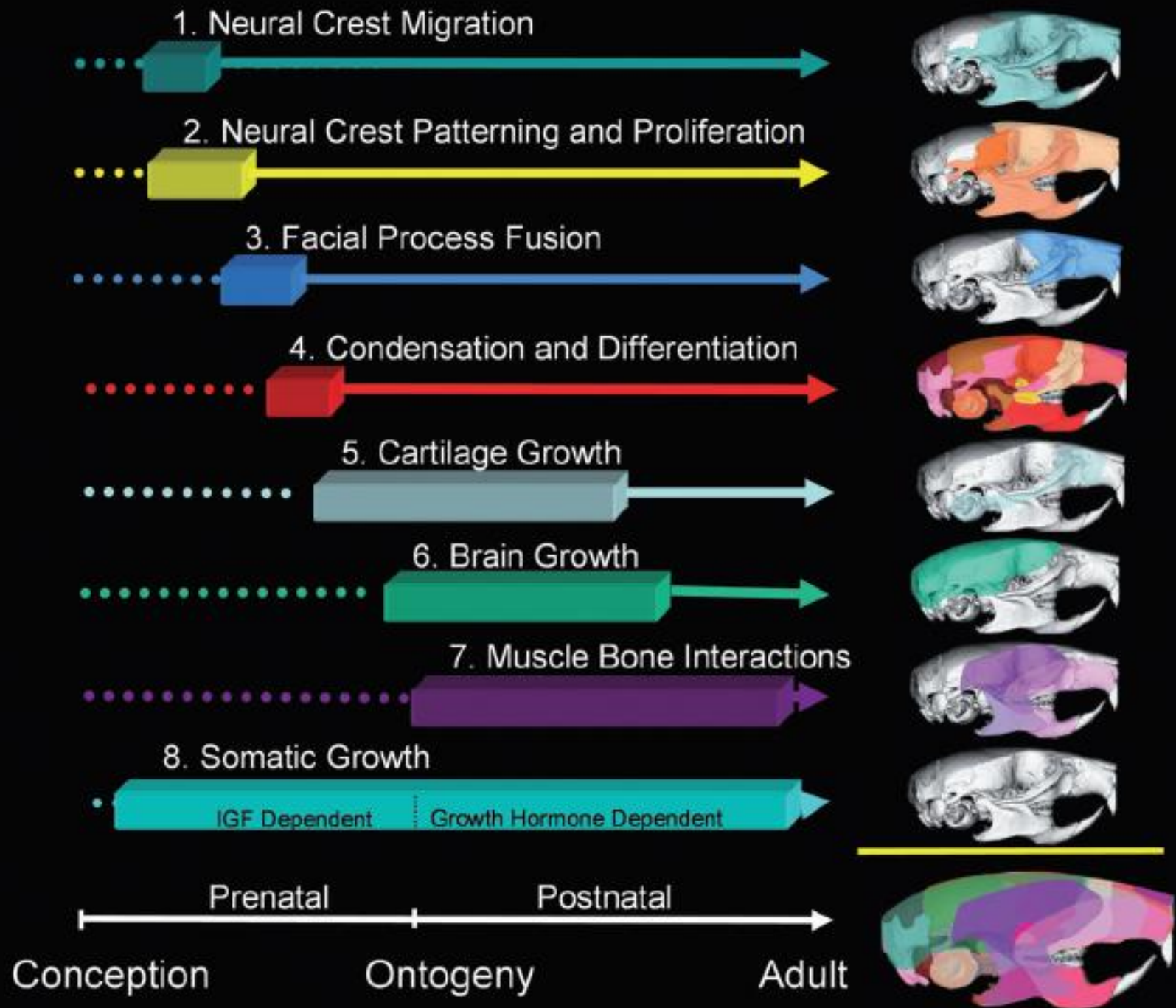
PROCESSOS
INDIVIDUAIS

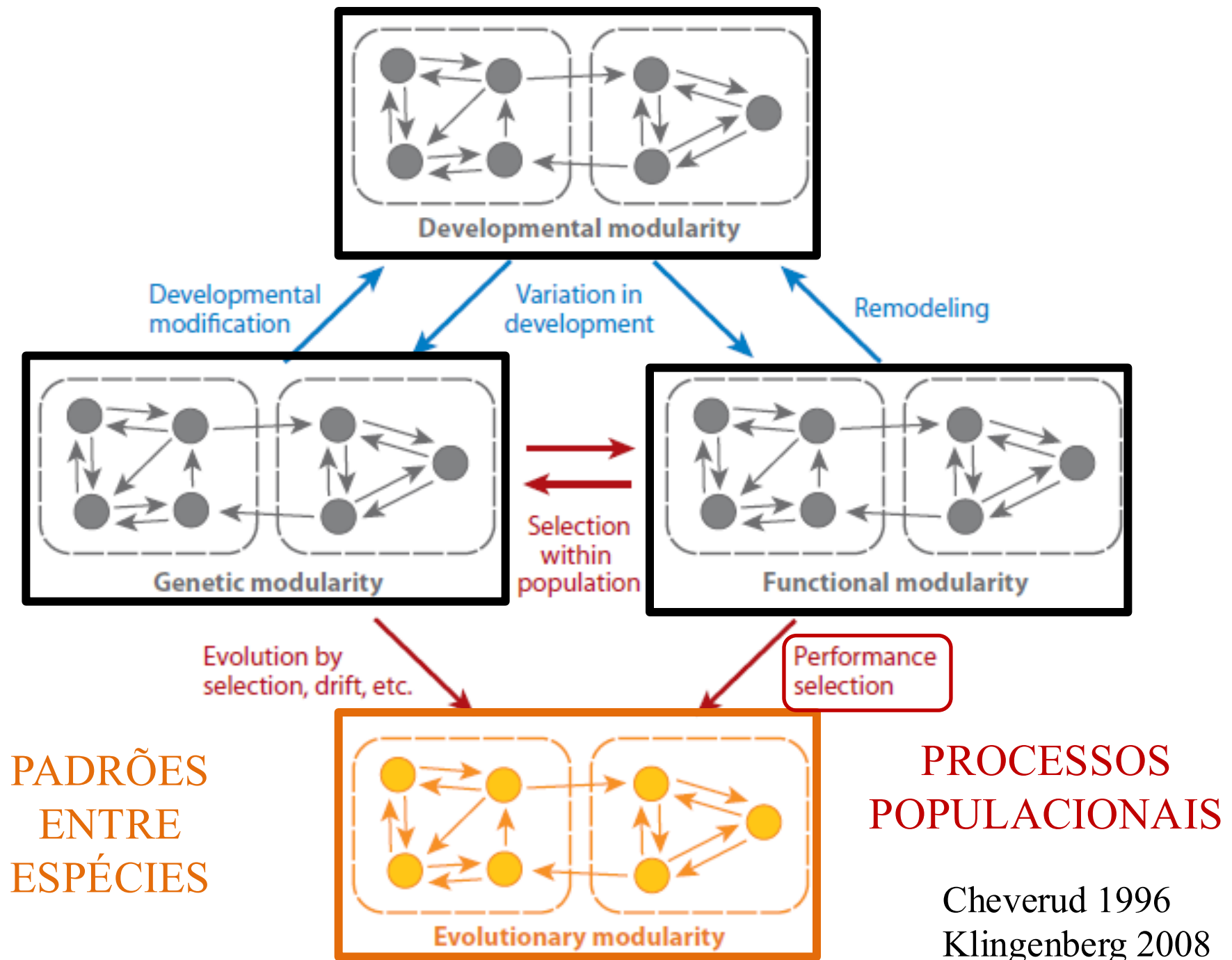
PADRÕES
DENTRO DE
POPULAÇÕES/
ESPÉCIES



PROCESSOS
POPULACIONAIS

Cheverud 1996
Klingenberg 2008

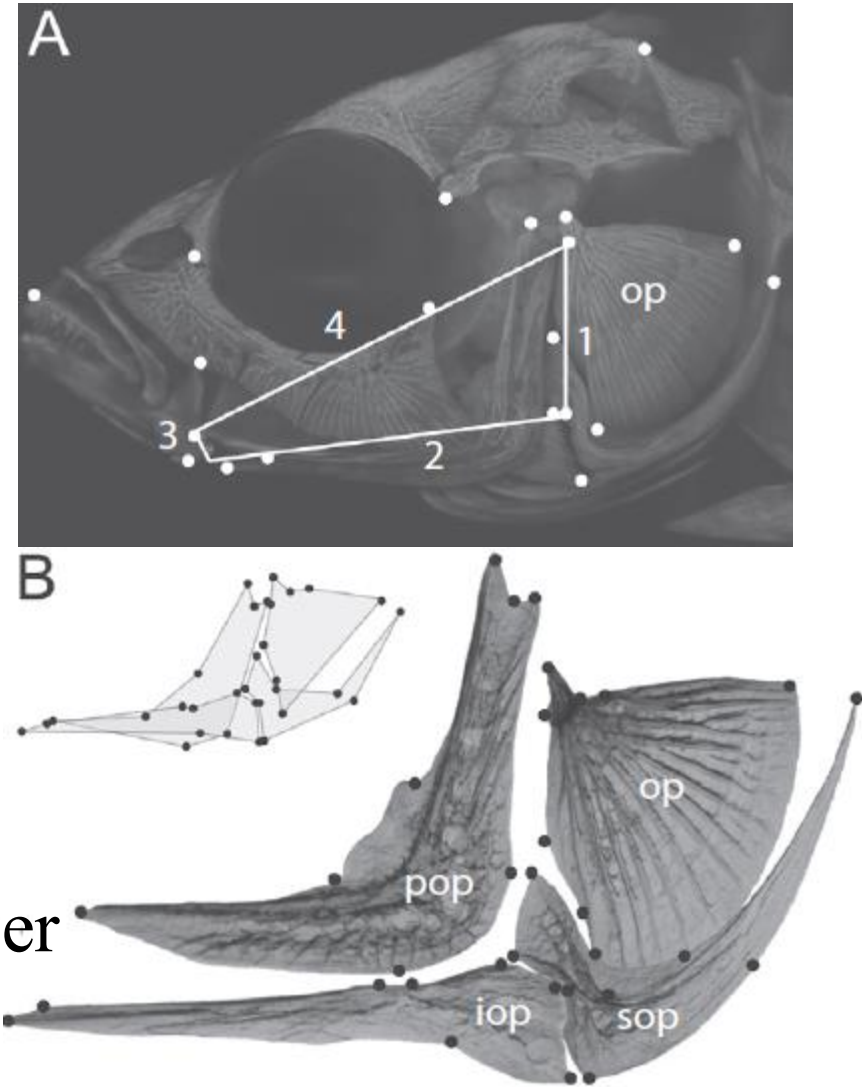




Seleção em desempenho: Modularidade funcional para evolutiva

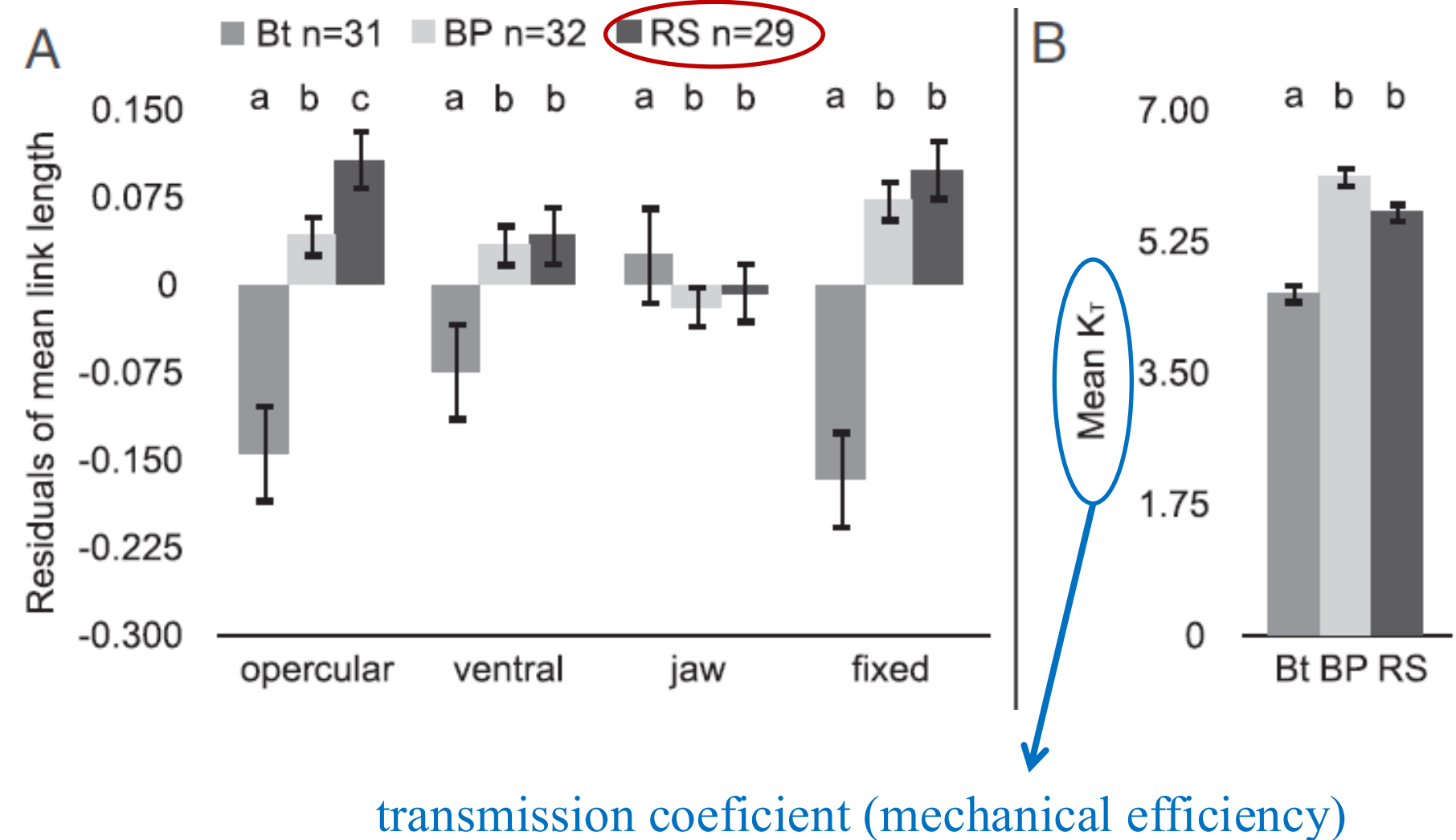


Opercular four bar lever



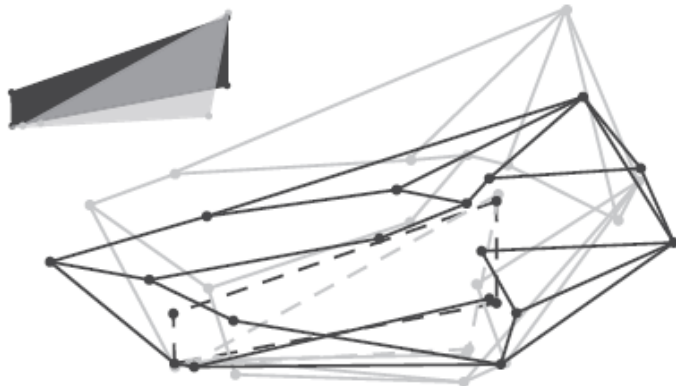
Diferenças biomecânicas na alavanca opercular de 4 elementos

marine ancestral

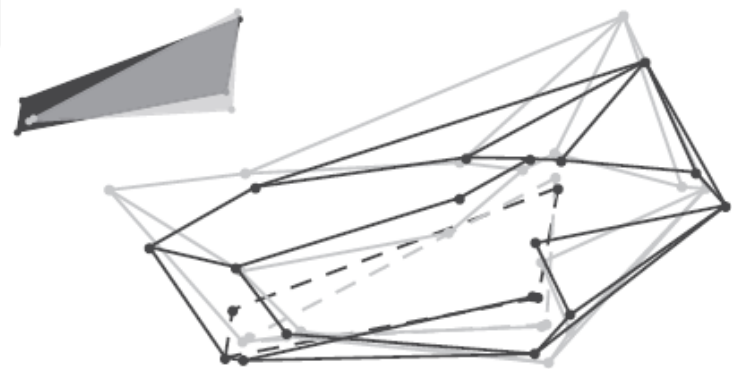


Variação na forma explica variação em K_T

A1

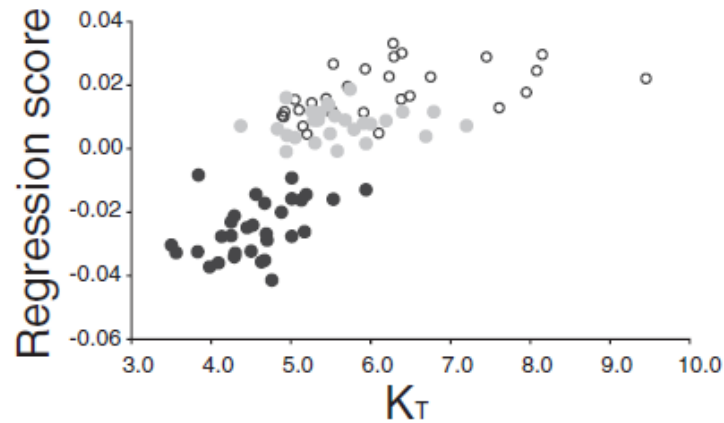


B1

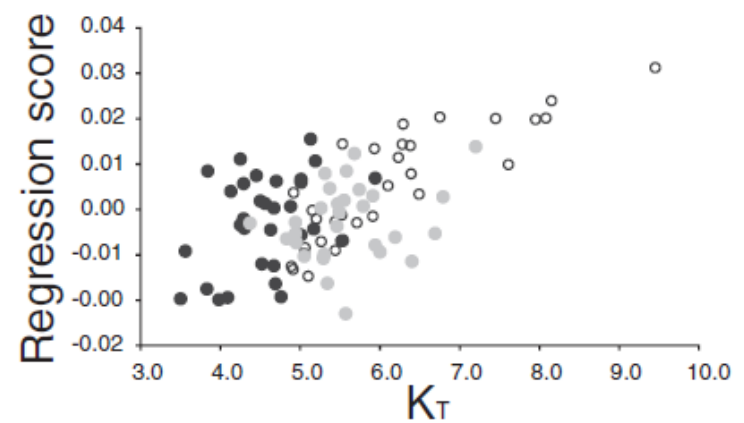


— (+)
— (-)

A2



B2



Bt BP RS
● ○ ●

Estabilidade do padrão e magnitude entre populações

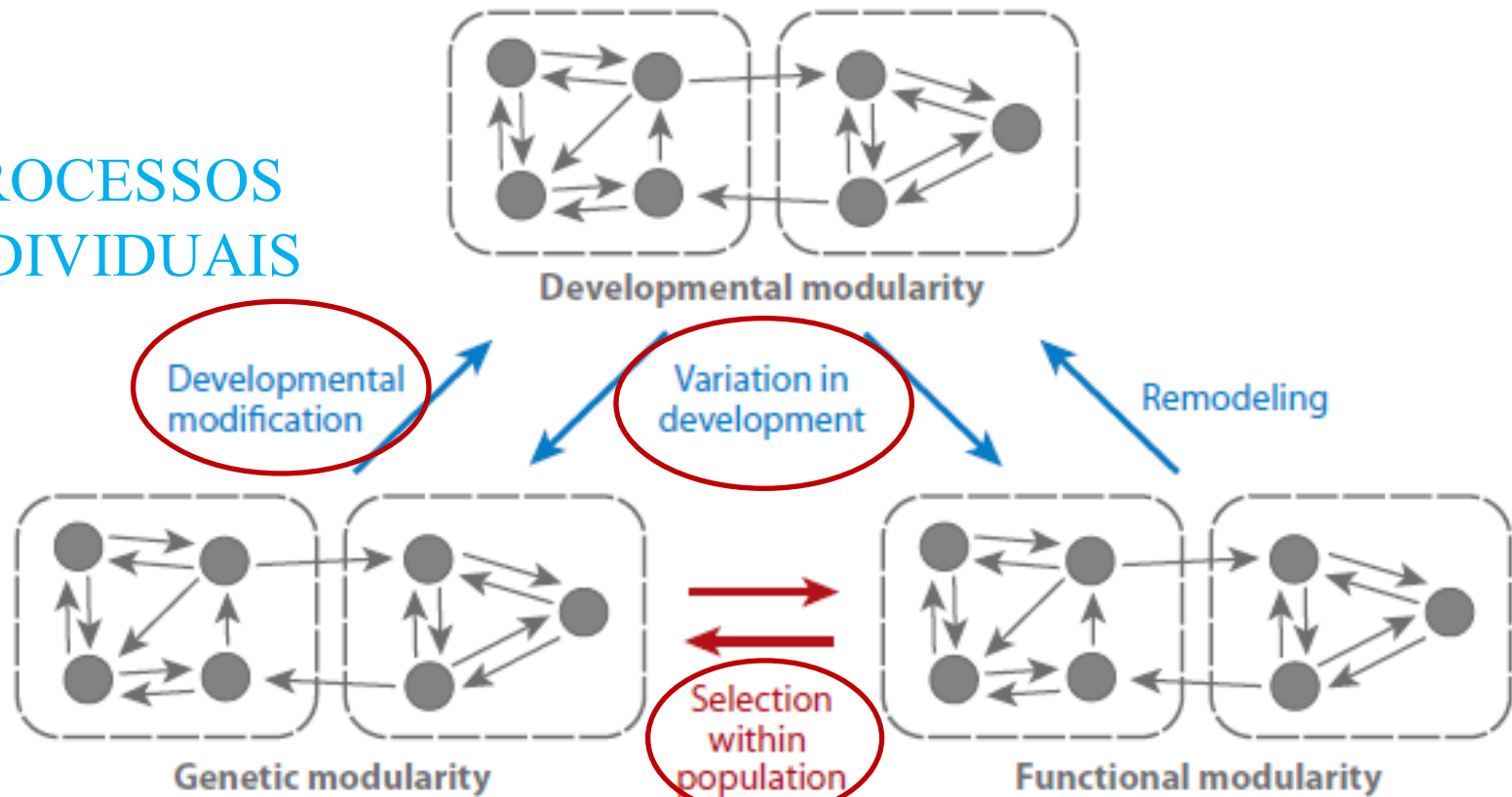
Table 1. Covariance matrix comparison among populations for the whole-skull dataset

	Boot Lake	Bear Paw Lake	Rabbit Slough
Boot Lake	0.8153	0.8817	0.9021
Bear Paw Lake	0.7366	0.8570	0.7931
Rabbit Slough	0.6491	0.5850	0.6349

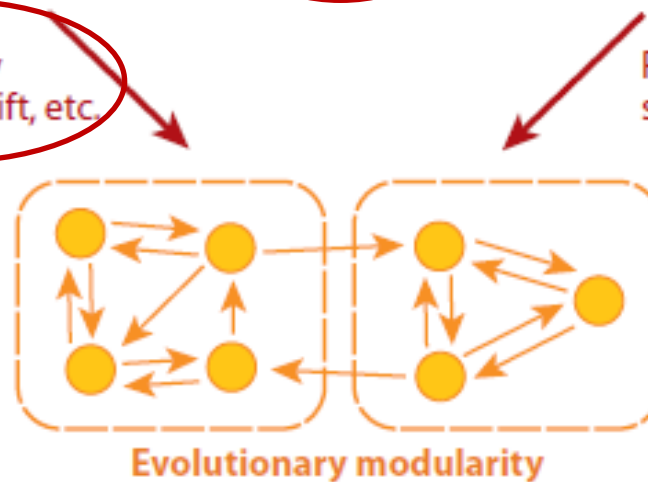
Population	Scaled variance of Eigenvalues	SD
Rabbit Slough	0.0012	0.0002
Boot Lake	0.0013	0.0001
Bear Paw Lake	0.0013	0.0002

As relações de forma e função podem mudar e suas médias fenotípicas mesmo que o padrão modular seja estável!

PROCESSOS INDIVIDUAIS



PADRÕES ENTRE ESPÉCIES



PROCESSOS POPULACIONAIS



Evolução por seleção: Modularidade funcional para genética para evolutiva



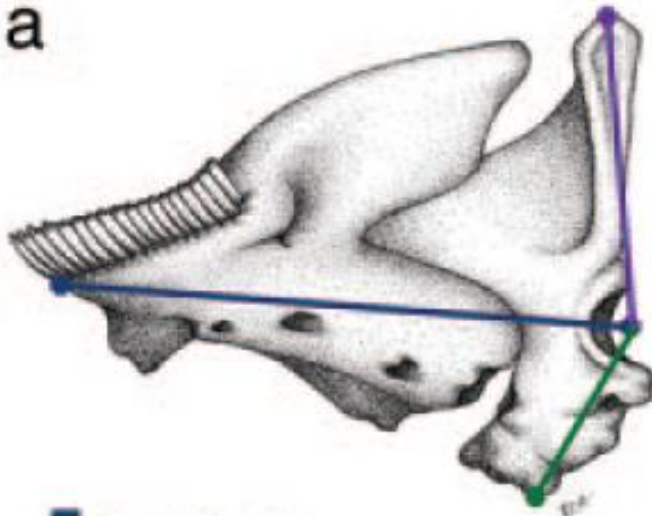
LF: Biting feeder



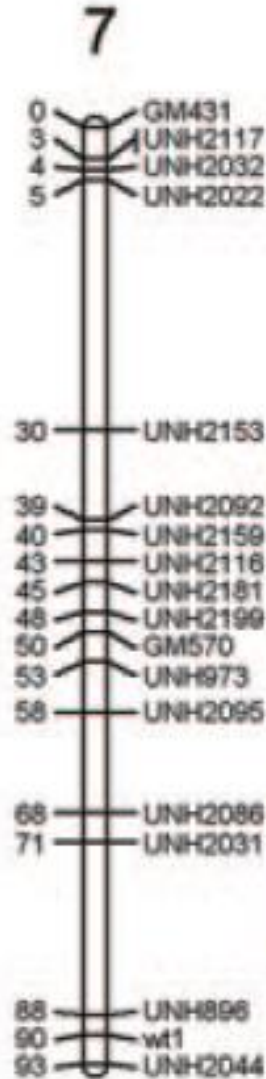
MZ: Suction feeder

Base genética para abertura e fechamento da boca

a



- Out-lever (L_o)
- Closing In-lever (L_{ci})
- Closing Mechanical Advantage ($MA_c = L_{ci}/L_o$)
- Opening In-lever (L_{oi})
- Opening Mechanical Advantage ($MA_o = L_{oi}/L_o$)



X

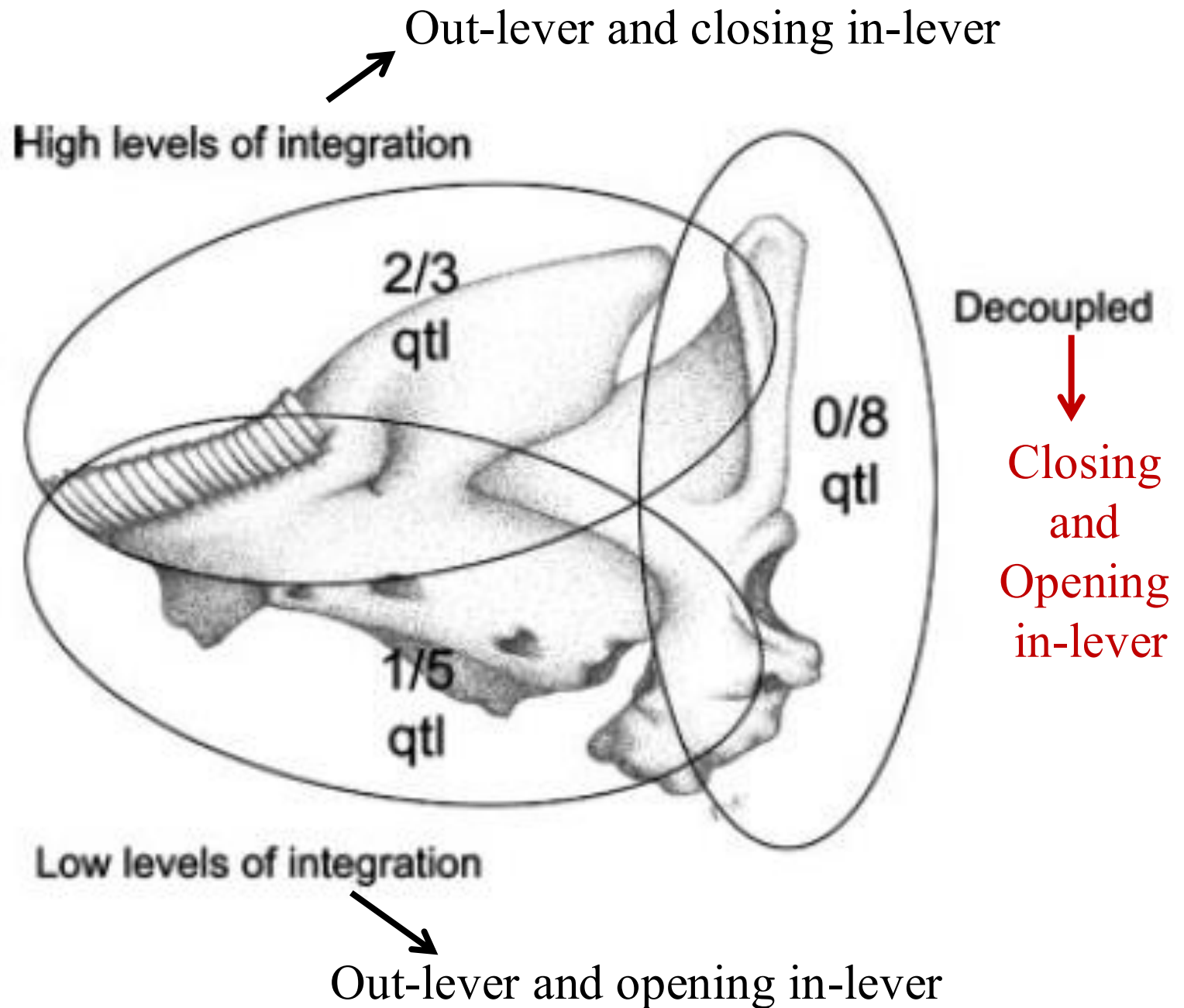


F2

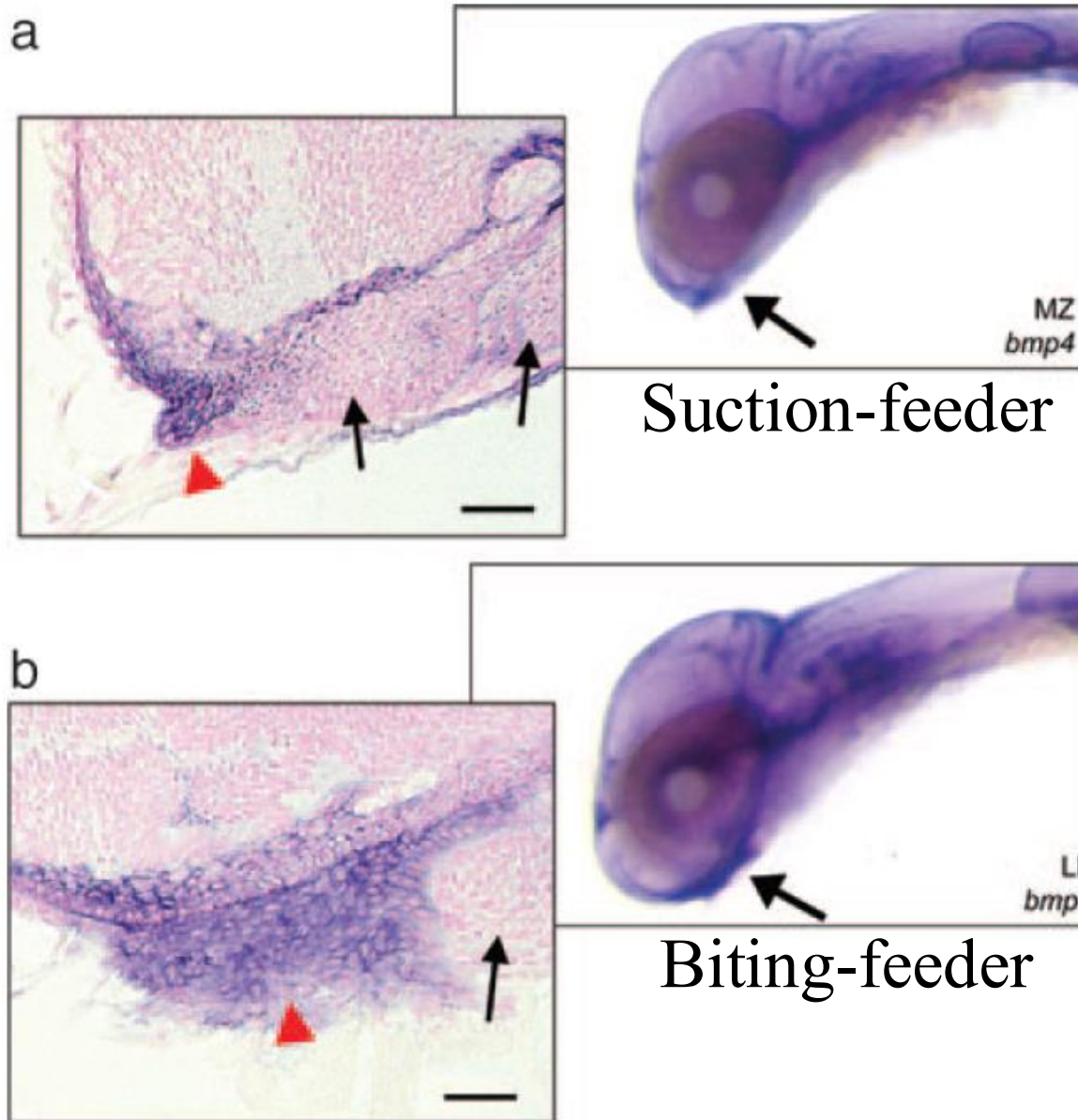
QTL mapping

Ossos de
abertura e

fechamento em
regiões distintas



Expressão divergente de *bmp4*



Divergência na biomecânica do forrageamento



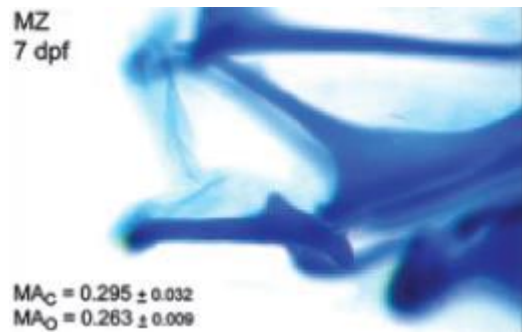
Seleção dentro
de populações

Desacoplamento genético de
módulos funcionais

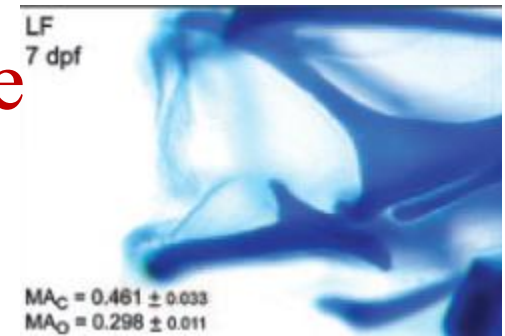


Modificação do
desenvolvimento

Expressão diferencial de *bmp4*



Diferenças entre
espécies



Evolução por seleção:

Modularidade funcional para desenvolvimento para evolutiva



Limb Development

Embryonic Patterning

Fetal/Postnatal Growth

Shared Factors

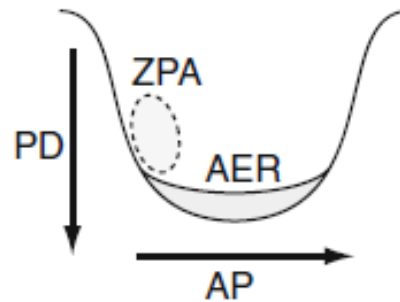
Genotype

Unique Factors

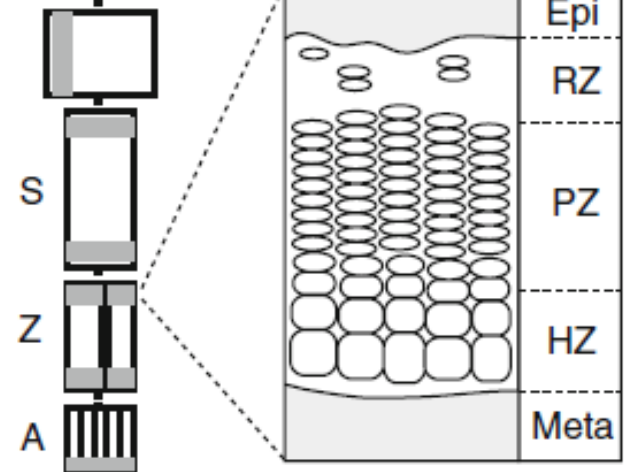
AP: Shh, HoxA, HoxD
PD: Shh, FGFs. S: HoxA10, D10.
Z: HoxA11, D11, A: HoxA13, D13 (etc)

Systemic factors: GH, IGF1 (etc)

1

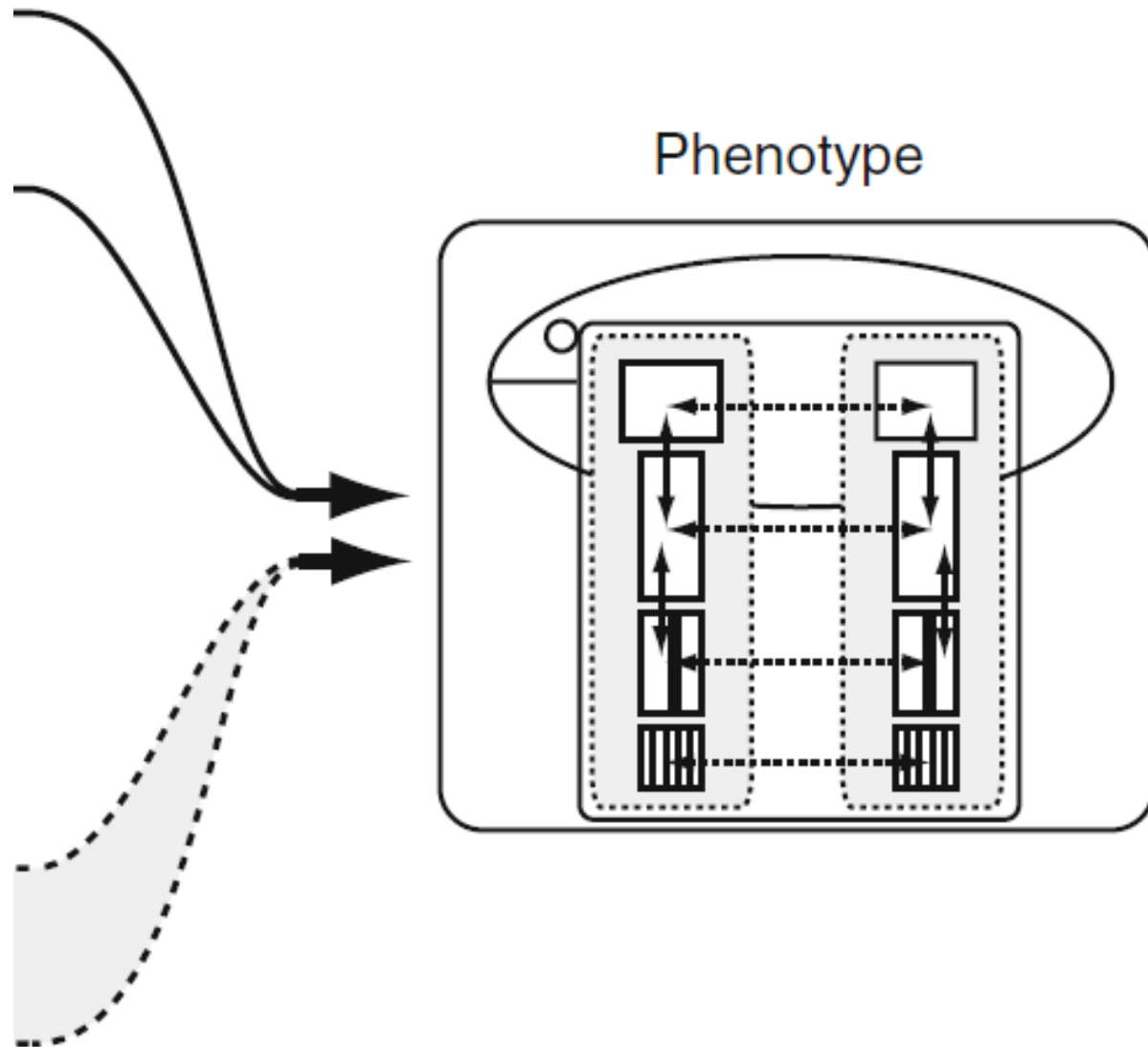


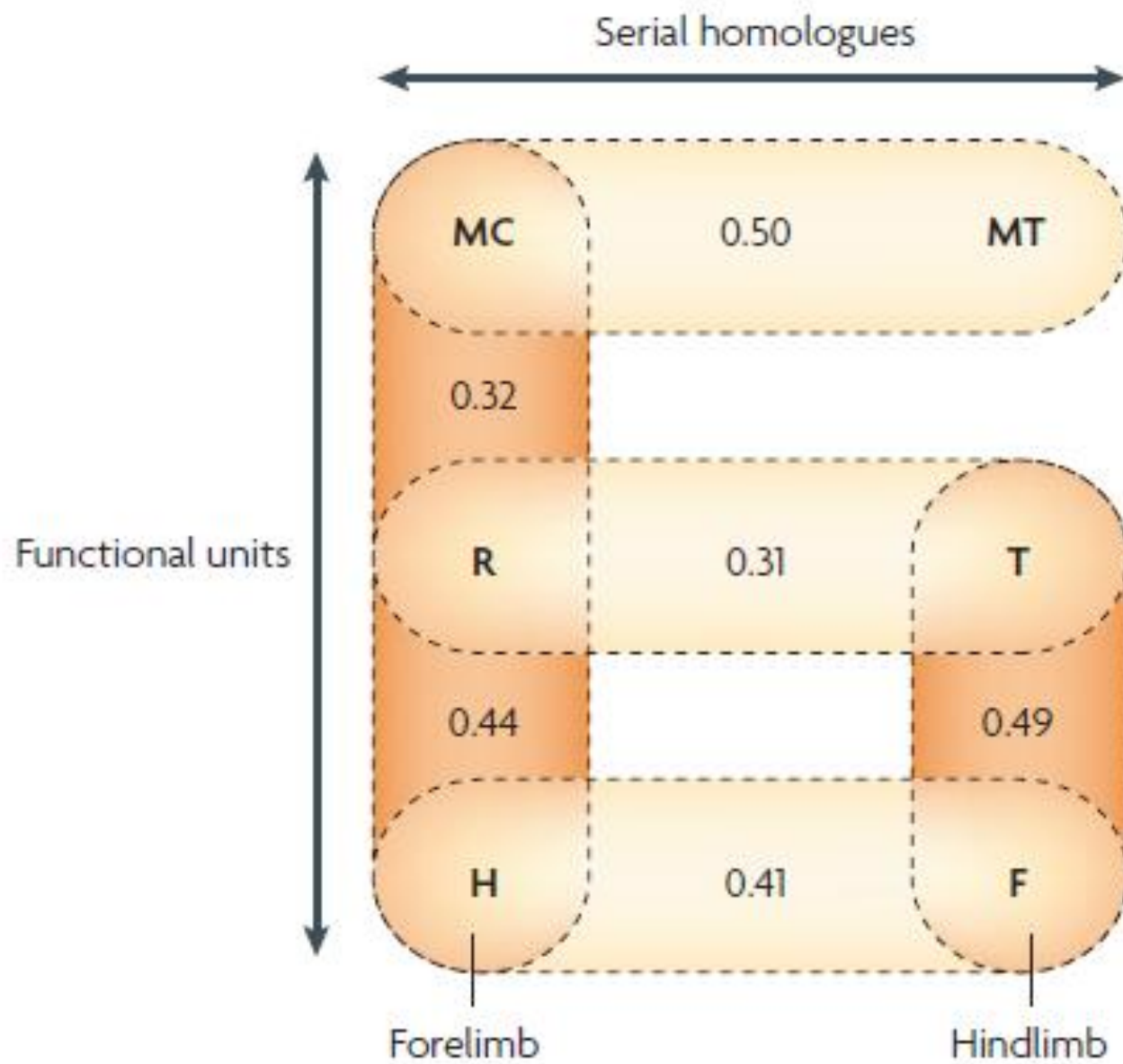
2



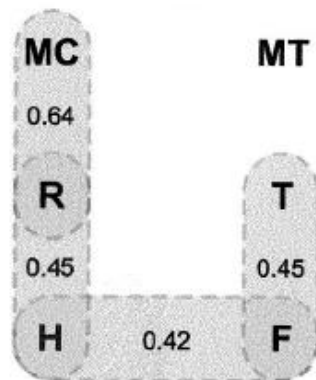
FL: Tbx5. HL: Pitx1, Tbx4
S: HoxA10, D10. Z: HoxA11, D11
A: HoxA13, D13 (etc)

Local factors: Ihh, PTHrP, IGF1 (etc)

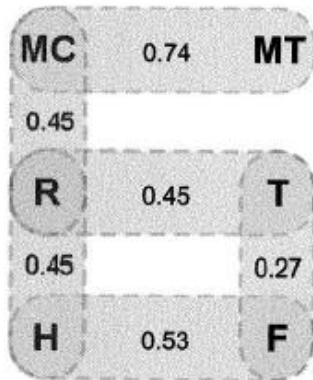




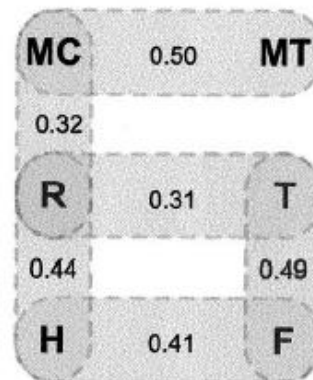
Bat



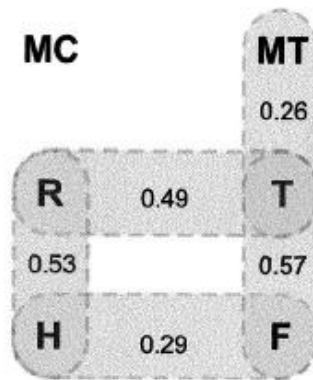
Gibbon



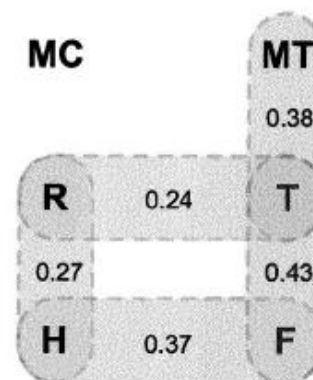
Macaque

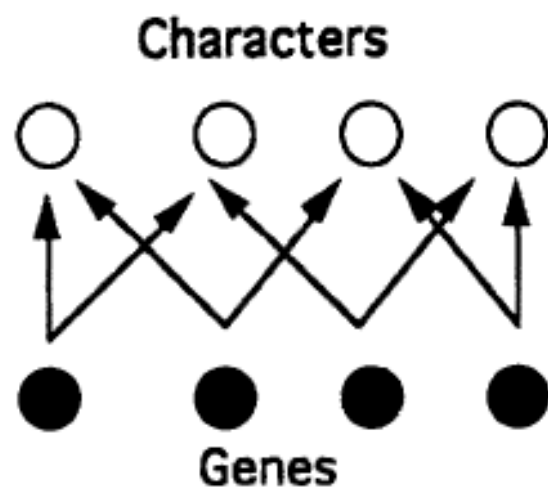


Lophuromys

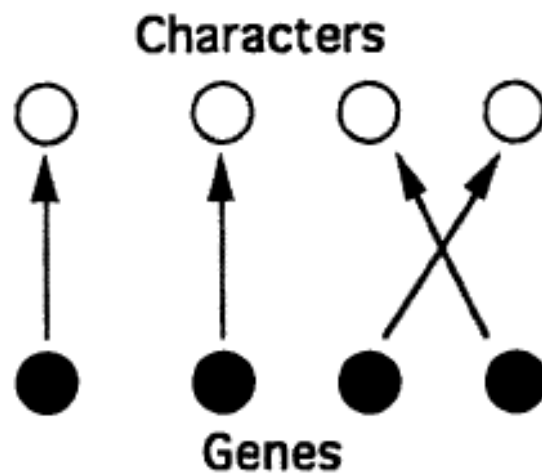
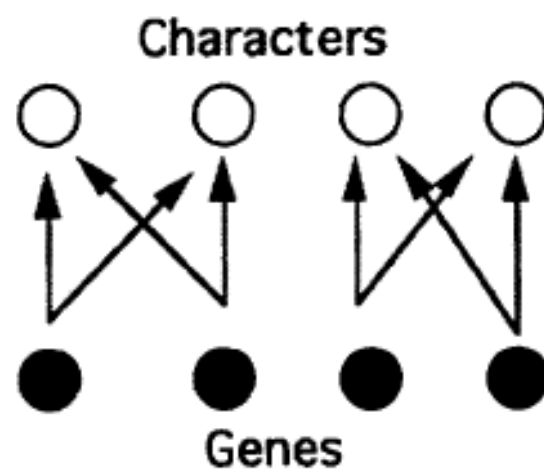


Mus





Parcellation

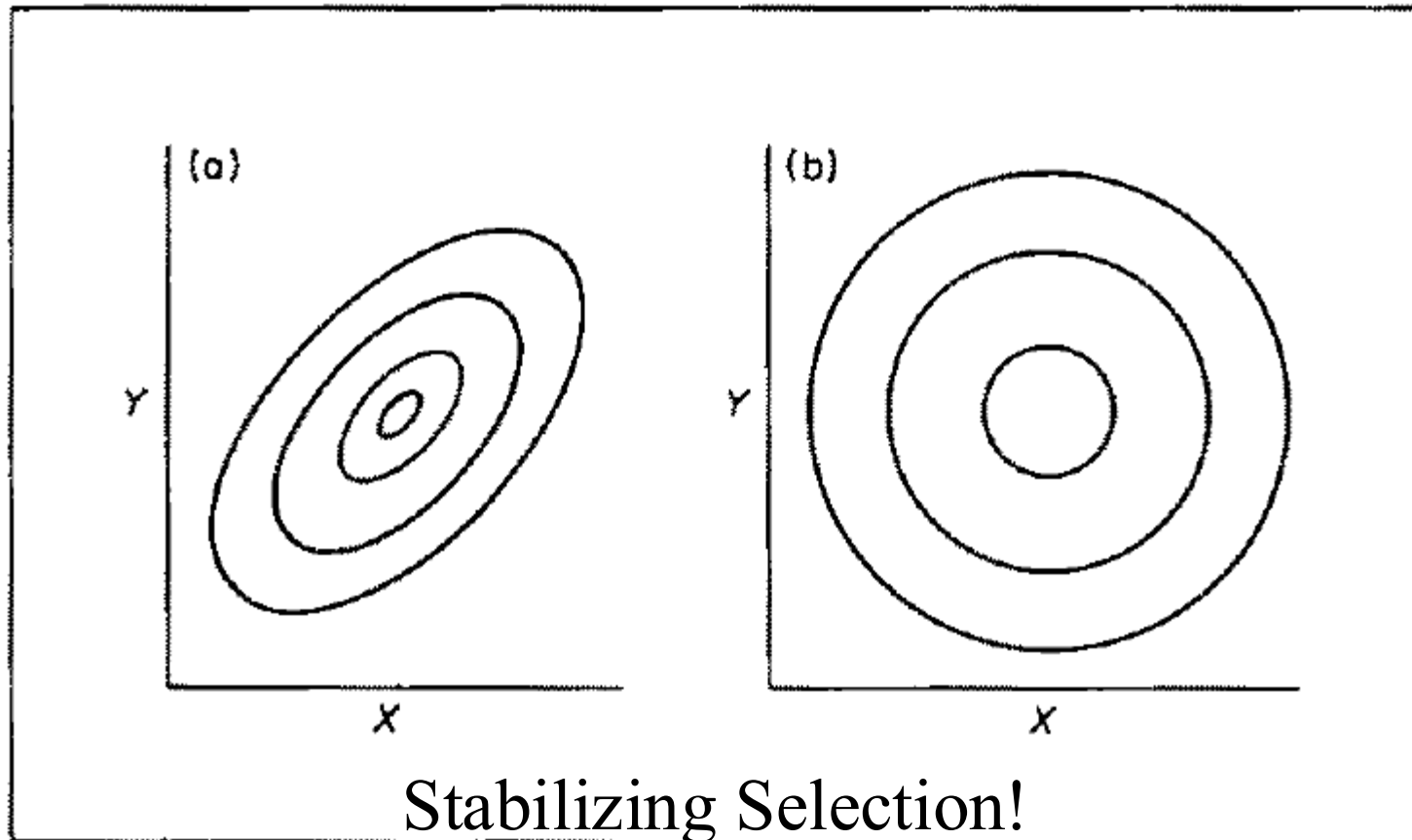


Integration

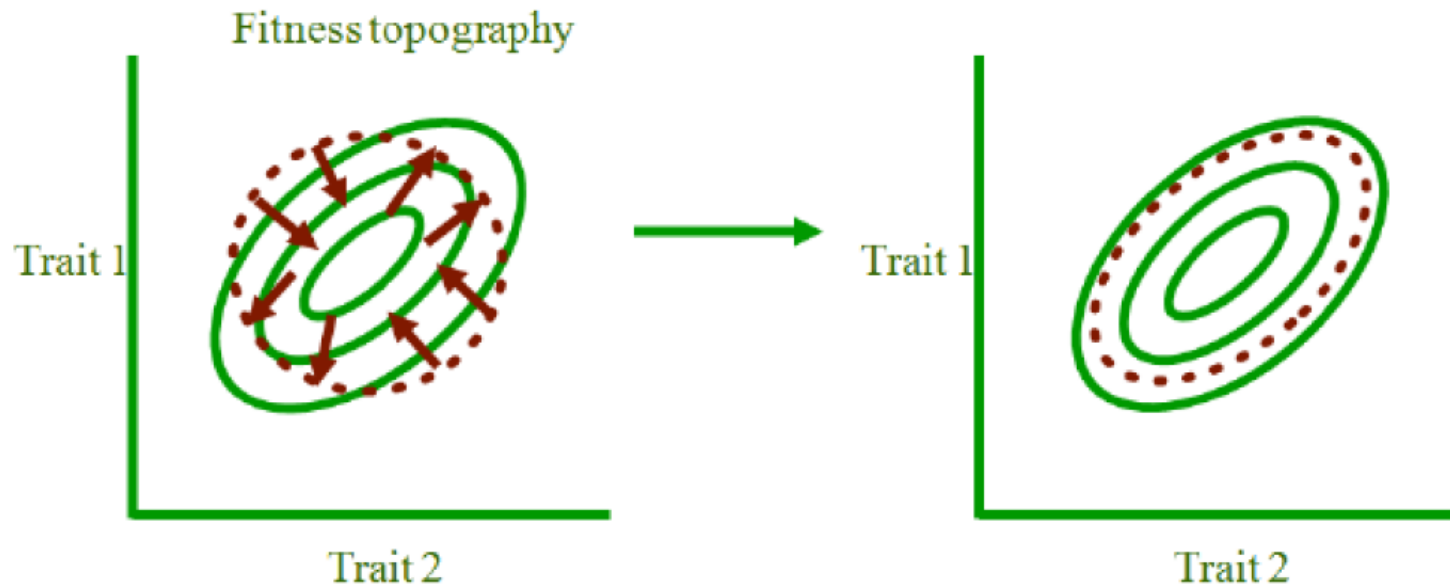
Quantitative Genetics and Developmental Constraints on Evolution by Selection

JAMES M. CHEVERUD

J. theor. Biol. (1984)

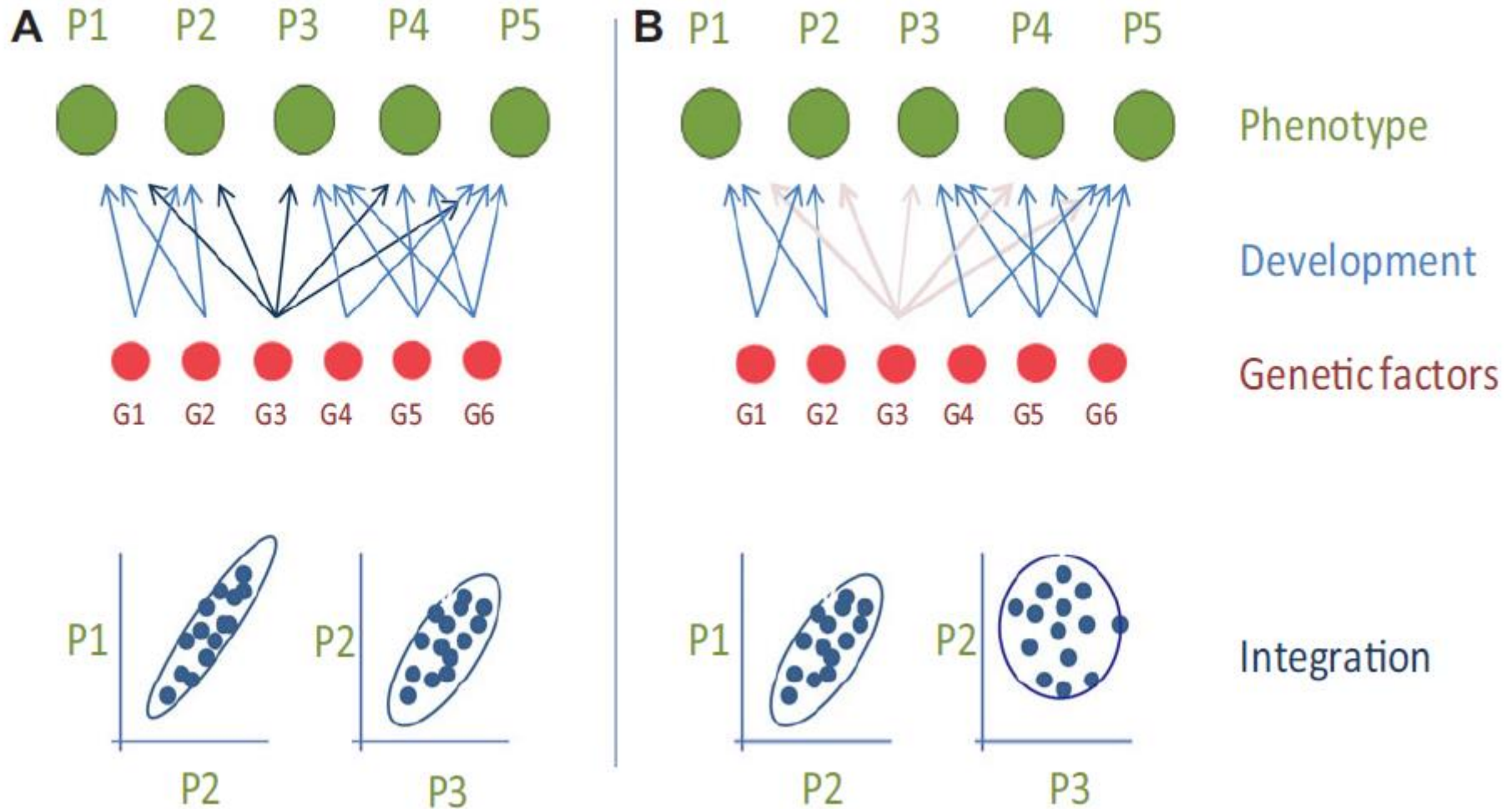


Quantitative genetics models: Riedl's hypothesis



Base genética do desenvolvimento é moldada por seleção estabilizadora multivariada (**correlational selection**) vinda de associações funcionais entre caracteres.

Mapa Genótipo-Fenótipo e Modularidade/Integração



E os efeitos ambientais?

